

Test Certificate

Product Code:	N3D11DEB012BK
Description:	Up to 24f , Axial loose tube, Quad FRP, Non-Metallic Armoured
Cable #:	12383/1
Date Cert created:	08 th September 2021
Design Number:	62-15
Product Family:	N3D1xxEx ⁺⁺⁺ XX

TEST PARAMETER / METHOD	AFL ACCEPTANCE CRITERIA	ACHIEVED	RESULT
Tensile Strength IEC 60794-1-21 (E1)	6kN 30min, < 0.6% fibre strain, No att. Increase after test, < 0.1dB during test, @ 1550nm	6kN <0.3% fs $\Delta\alpha \leq 0.06\text{dB}$ No att. Increase after test	Pass
Bend under tension IEC 60794-1-21 (E18A)	$r_b = 20 \times \text{OD}$ 360°, 1 turn, 1 min No fibre break, no damage to sheath or core structure and No attn. change throughout test	$r_b = <20 \times \text{OD}_{\text{measured}}$ No breaks or damage and no attenuation change	Pass
Crush (short term) IEC 60794-1-21 (E3)	6kN/100mm Duration: 10min 3 position No damage to sheath or core structure and No att. Increase after test	6kN/100mm No damage and no attenuation increase	Pass
Crush (long term) IEC 60794-1-21 (E3)	4kN/100mm Duration: 120min 3 positions No damage to sheath or core structure and No att. Increase during or after test	4kN/100mm No damage and no attenuation increase	Pass
Impact IEC 60794-1-21 (E4)	Weight: 1.5kg Height: 1.0m, 300mm rad anvil, 3 impacts, After 5 mins No fibre break, no damage to sheath or core structure and No attn. change throughout test	No breaks or damage and no attenuation change	Pass
Bend IEC 60794-1-21 (E11)	$r_b = 10 \times \text{OD}$ 360° loop, 1 turn, 1 cycle No att. Change throughout test	No attn. Change	Pass
Torsion IEC 60794-1-21-(E7)	Sample length: 1 m Rotation: a) 180° clockwise, b) return to starting position, c) 180° anticlockwise, d) return to starting position. Four movements = one cycle. Complete each cycle within 1 min max. for total 10 cycles (a to d) During the final tenth cycle at a), c) and after completion (no rotation) check transmitting fibres. No fibre breaks, no damage to the sheath or to the core structure and no attenuation change throughout test	No breaks or damage and no attenuation change	Pass
Repeat Bending IEC 60794-1-21-(E6)	No. of cycles: 20 Weight: 4kg $r_b = 7 \times \text{OD}$ Ambient temp: 20.6°C 90° bend, return to centre starting position, 90° bend opp. direction return to centre starting position within 2-5s = one cycle	No breaks or damage and no attenuation change	Pass
Temperature Cycle IEC 60794-1-22 (F1)	Temperature Range: -40°C to +70°C $\Delta\alpha_{\text{max}} \leq 0.15\text{dB}$ $\Delta\alpha_{\text{avg}} < 0.05\text{dB}$ @ 1550nm	Temperature Range: -40°C to +70°C $\Delta\alpha_{\text{max}} \leq 0.15\text{dB}$ $\Delta\alpha_{\text{avg}} < 0.05\text{dB}$ @ 1550nm	Pass
Water Penetration IEC 60794-1-22 (F5B)	Sample length: 3m Water Height: 1m No water after 24 hours	No Water detected	Pass
Tested By:	Attila & Karlo	Verified By:	Shashini
		Approved By:	Jim Boukouvalas